

ASCII MASTER FLOW – PANEL 1/12: Karst coupled system

RECHARGE -> joints, bedding and sink points

DISSOLUTION -> openings enlarge underground

CONDUITS -> springs export water and solute

SURFACE + CAVE -> one hydrological landscape

MUST REMEMBER: Karst requires soluble rock, joints, water and carbonic-acid solution, followed by underground drainage and carbonate deposition; lapies, sinkholes, dolines, uvalas, poljes, caves, stalactites and stalagmites form a connected hydrological system.

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ASCII MASTER FLOW – PANEL 2/12: Carbonate cycle

CO₂ + WATER -> weak carbonic acid

ACID + CaCO₃ -> dissolved bicarbonate

TRANSPORT -> solution moves through conduits

DEGASSING -> calcite can precipitate

ASCII MASTER FLOW – PANEL 3/12: Karst prerequisites

ROCK -> soluble, thick and exposed to recharge

STRUCTURE -> joints and bedding pathways

FLOW -> gradient plus outlet / base level

TIME -> sustained dissolution and removal

ASCII MASTER FLOW – PANEL 4/12: Hydrological zones

EPKARST -> shallow storage and routing

VADOSE -> descending water in air-filled voids

WATER TABLE -> fluctuating boundary

PHREATIC -> saturated tubes and loops

ASCII MASTER FLOW – PANEL 5/12: Surface-karst ladder

KARREN -> small solution grooves

DOLINE -> closed depression

UVALA / POLJE -> larger composite / structural basin

SWALLOW HOLE -> surface stream transfers underground

ASCII MASTER FLOW – PANEL 6/12: Cave-development sequence

INCEPTION -> favourable fracture or bedding plane

PHREATIC -> passage enlarges under saturation

VADOSE -> stream incises after water-table fall

LATE CHANGE -> collapse, fill or renewed solution

ASCII MASTER FLOW – PANEL 7/12: Speleothem map

STALACTITE -> roof drip deposit

STALAGMITE -> floor splash / drip deposit

COLUMN -> roof and floor forms join

FLOWSTONE / CURTAIN -> film flow on wall or slope

ASCII MASTER FLOW – PANEL 8/12: India cave firewall

MAWMLUH / MAWSMAI / SIJU -> Meghalaya limestone

BORRA / BELUM -> natural caves in Andhra Pradesh

AJANTA / ELLORA -> human rock-cut architecture

RULE -> location does not replace genetic classification

ASCII MASTER FLOW – PANEL 9/12: Proxy pipeline

CLIMATE -> rainfall, vegetation and soil CO₂

RECHARGE -> route and residence time

CALCITE -> isotope / element signal

CHRONOLOGY -> dated series with uncertainty

ASCII MASTER FLOW – PANEL 10/12: Meghalayan hierarchy

QUATERNARY -> Period

HOLOCENE -> Epoch

MEGHALAYAN -> Age / Stage

GSSP -> KM-A speleothem at Mawmluh

CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the Meghalayan chronostratigraphic unit are categorically different.

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ASCII MASTER FLOW – PANEL 11/12: Status and narrative caution

4.2 ka EVENT -> correlation guide

GSSP POINT -> physical lower boundary

MEGHALAYAN -> named formal interval

ANTHROPOCENE -> not a ratified Epoch

EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary
must not be collapsed; tourism and infrastructure effects depend on recharge, passage
geometry, biodiversity and carrying capacity.

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ASCII MASTER FLOW – PANEL 12/12: Karst conservation spine

MAP -> recharge, passages, springs and values

AVOID -> irreversible quarry and pollution pathways

MANAGE -> visitation, waste and water use

MONITOR -> hydrology, biodiversity and structural change